Lecture 3:-

Basics of Statistics, Data Type, One-Hot Vectors

Statistics 101:-

"Academic discipline dealing with all aspects of (learning from) data (quantification)".

Perspectives (why need):-

- art of summarizing data. → make data comprehensible.
- science of uncertainty → most information in the world is uncertain.
- science of decisions. → ultimate goal of statistics.
- science of variation. → central tendency and spread.
- art of forecasting.
- science of measurement and data collection.

Foundations of data:-

Source of data:-

- Designed data – "artificially collected" (surveys, studies etc)
- Organic data (process generated)

Note:- Organic data is widely used in computer science in statistics

For both, data needs to be i.i.d "independent", "identically distributed". → discuss more on this later!

Types of data :-

- Some data is not numeric !
  For instance race or gender.
- Just as we have data types in programming languages,

We have different types here
- Weight — numeric, continuous
- Number of kids — numeric, discrete.
- Age group (child, adult, elder) — categorical, ordered
- Gender — categorical, unordered

Practical Note :

Gender represented as : M / F

or : 0 / 1 → But still ordered!

or : $\begin{bmatrix}1\\0\end{bmatrix} / \begin{bmatrix}0\\1\end{bmatrix}$ → now unordered!

"One-hot vector representation".

For example

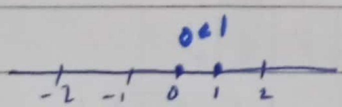
$B=0, W=1, H=2$

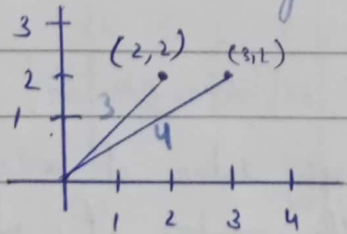
$$\begin{array}{c|c}
\text{index} & \\
0 \\
1 \\
2
\end{array}
\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$$

B, W, H

One-hot vector :-

In One-hot vector the elements are no longer compare able the reson is given below



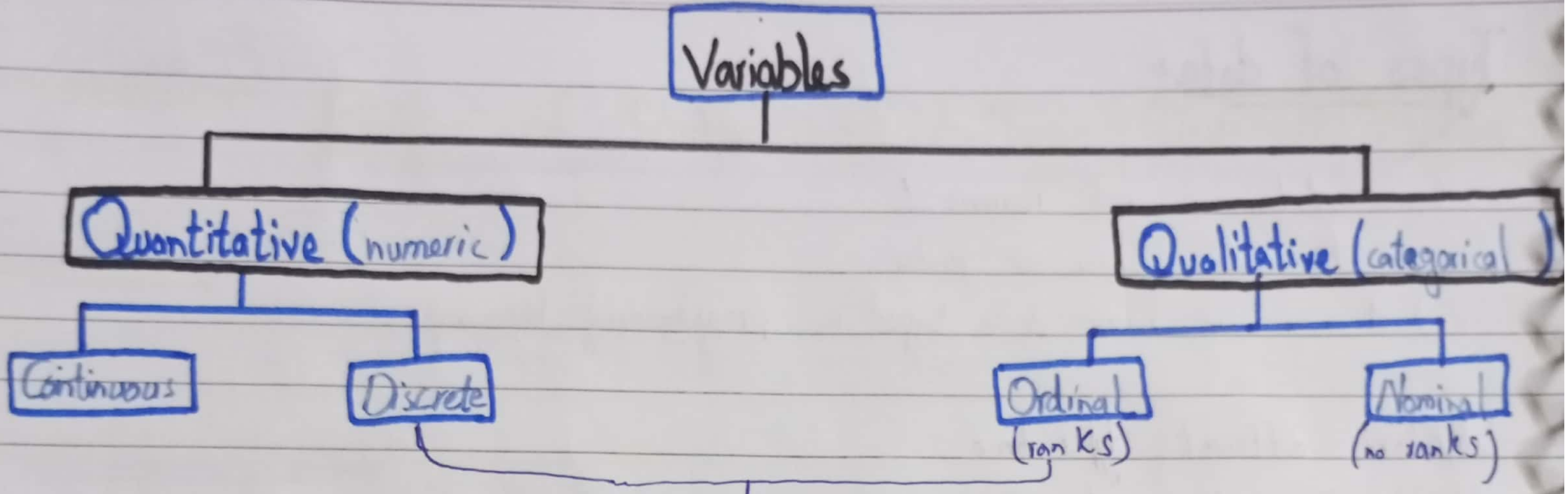


Magnitude :- $3 < 4$ (True compare able)

But :-

$(2,2) < (3,2)$ → uncomparable

As they have two values on both sides so they are no longer compareable



Both discrete & ordered!
So difference?
Ans:- In discrete averages make sense while in ordinal averages not make sense.

The further detail is written in file or practical are done in file "01-data-types.ipynb".